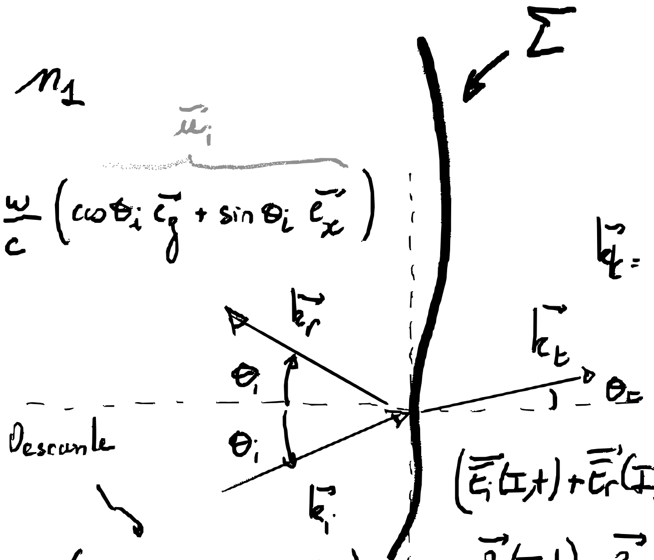
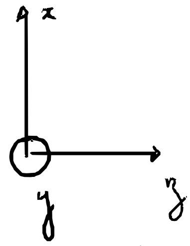
Explication TP He-Ne

I. Coefficients de réflexion et transmission





$$\vec{k_i} = n_1 \frac{\omega}{c}\overbrace{\left(\cos\theta_i \vec{e_z} + \sin\theta_i \vec{e_x}\right)}^{\vec{u_i}}$$

$$\vec{k_t} = n_2 \frac{\omega}{c}\underbrace{\left(\cos\theta_t \vec{e_z} + \sin\theta_t \vec{e_x}\right)}_{\vec{u_t}}$$

Premier loi de Descarte

$$\vec{k_r} = n_1 \frac{\omega}{c}\underbrace{\left(-\cos\theta_i \vec{e_z} + \sin\theta_i \vec{e_x}\right)}_{\vec{u_r}}$$

$$\left(\vec{E_i}(I,t) + \vec{E_r}(I,t)\right)_{//} = \vec{E_t}(I,t)_{//} \quad (1)$$

$$\vec{B_i}(I,t) + \vec{B_r}(I,t) = \vec{B_t}(I,t) \ ; \ \forall I \in \Sigma$$

or pour des ondes planes $\vec{B} = \frac{\vec{k}\wedge\vec{E}}{\omega} = \frac{n}{c}\vec{u}\wedge\vec{E}$

donc : $n_1 \vec{u_i}\wedge\vec{E_i} + n_1 \vec{u_r}\wedge\vec{E_r} = n_2 \vec{u_t}\wedge\vec{E_t}$

$$n_1\left(\cos\theta_i \vec{e_z}\wedge\left(\vec{E_i}(I,t) - \vec{E_r}(I,t)\right) + \sin\theta_i \vec{e_x}\wedge\left(\vec{E_i}(I,t) + \vec{E_r}(I,t)\right)\right) =$$
$$n_2\left(\cos\theta_t \vec{e_z}\wedge\vec{E_t}(I,t) + \sin\theta_t \vec{e_x}\wedge\vec{E_t}(I,t)\right)$$

$$\rightarrow \left(\begin{array}{l} \vec{e_z}\wedge\left[n_1\cos\theta_i\left(\vec{E_i}(I,t) - \vec{E_r}(I,t)\right) - n_2\cos\theta_t \vec{E_t}(I,t)\right] \\ + \vec{e_x}\wedge\left[n_1\sin\theta_i\left(\vec{E_i}(I,t) + \vec{E_r}(I,t)\right) - n_2\sin\theta_t\vec{E_t}(I,t)\right] = 0 \end{array}\right\} (**)$$

On regarde dans le plan $\perp$ à $\vec{e_z}$ (tangencielle) on fait $\vec{e_z}\wedge$ à $(**)$ et $\vec{a}\wedge(\vec{b}\wedge\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$

$$\rightsquigarrow \left(\left[n_1\cos\theta_i\left(\vec{E_i}(I,t) - \vec{E_r}(I,t)\right) - n_2\cos\theta_t\vec{E_t}(I,t)\right]\cdot\vec{e_z}\right)\vec{e_z}$$
$$- \left[n_1\cos\theta_i\left(\vec{E_i}(I,t) - \vec{E_r}(I,t)\right) - n_2\cos\theta_t\vec{E_t}(I,t)\right]$$
$$+ \left(\left[n_1\sin\theta_i\left(\vec{E_i}(I,t) + \vec{E_r}(I,t)\right) - n_2\sin\theta_t\vec{E_t}(I,t)\right]\cdot\vec{e_z}\right)\vec{e_x} = 0$$

(transverses magnétique)

$$+\left(\left[n_2 \sin\theta_i \left(\vec{E}_i(I,t)+\vec{E}_r(I,t)\right)-n_2 \sin\theta_t \vec{E}_t(I,t)\right]\cdot\vec{e_z}\right)\vec{e_x} = 0$$

(transverses magnétique)

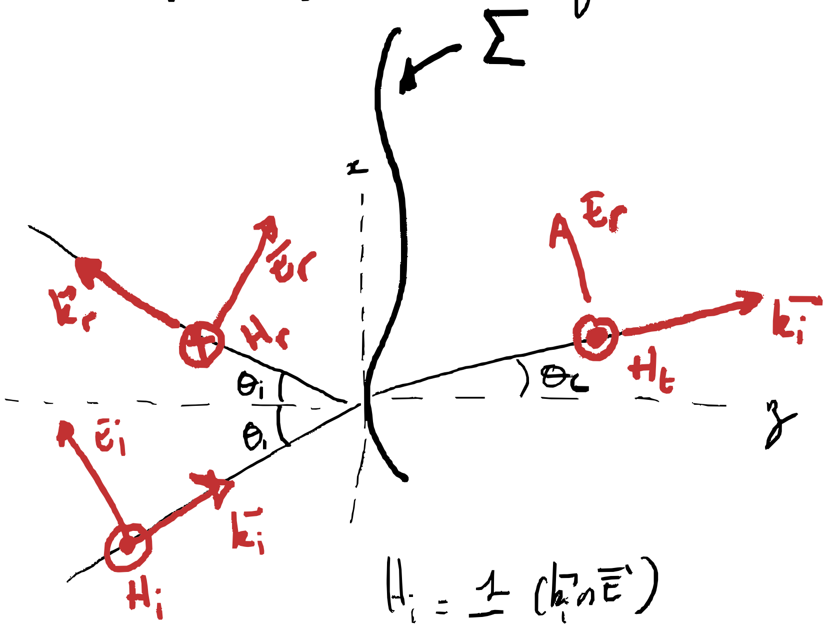
Polarisation p : Le champ électrique est dans le plan incidence (x,z) Donc les composantes tangentielles nécessitent des projection en $\cos\theta$.

(*) devient $\left(\vec{E}_i(I,t)+\vec{E}_r(I,t)\right)\cos\theta_i = \vec{E}_t(I,t)\cos\theta_t \quad \forall I\in\Sigma$

or $\vec{E}_r(I,t) = r_p \vec{E}_i(I,t)$ et $\vec{E}_t(I,t) = t_p \vec{E}_i(I,t)$

soit $(1-r_p)\cos\theta_i = t_p \cos\theta_t$

et $\vec{E}\perp\vec{k}$ et $\vec{E}\in(x,z)$ donc $B \parallel y$ soit



$$H_i = \frac{1}{\mu_0\omega}(\vec{k_i}\wedge\vec{E})$$

$$\begin{cases} H_i - H_r = H_t \\ (E_i+E_r)\cos\theta_i = E_t\cos\theta_t \end{cases} \forall I\in\Sigma \Rightarrow \begin{cases} k_1(1-r_{TM}) = k_2 t_{TM} \\ (1+r_{TM})\cos\theta_i = t_{TM}\cos\theta_2 \end{cases}$$

d'où avec $k = n\,\omega/c$

$$r_{TM} = \frac{n_1\cos\theta_t - n_2\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_i} \quad \text{et} \quad t_{TM} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_t + n_2\cos\theta_1}$$

$n_1\sin\theta_i = n_2\sin\theta_t$

$$r_{TM} = 0 \Leftarrow n_2^2\cos^2\theta_1 = n_1^2\cos^2\theta_t = n_1^2(1-\sin^2\theta_t) = n_1^2\left(1-\frac{n_1^2}{n_2^2}\sin^2\theta_i\right)$$

$$\left(\frac{n_2^2}{n_1^2}\cos^2\theta_i = 1-\frac{n_1^2}{n_2^2}\sin^2\theta_i\right)\Big/\cos^2\theta_i \qquad \left\{\frac{1}{\cos^2\theta_i} = 1+\tan^2\theta_i\right\}$$

$$\frac{n_2^2}{n_1^2} = (1+\tan^2\theta_i) - \frac{n_1^2}{n_2^2}\tan^2\theta_i$$

$$\left(\frac{n_2^2}{n_1^2}\cos^2\theta_i = 1-\frac{n_1^2}{n_2^2}\sin^2\theta_i \Big/ \Big/ \cos^2\theta_i \qquad \left\{\frac{1}{\cos^2\theta_i} = 1+\tan^2\theta_i\right)\right.$$

$$\frac{n_2^2}{n_1^2} = \left(1+\tan^2\theta_i\right) - \frac{n_1^2}{n_2^2}\tan^2\theta_i$$

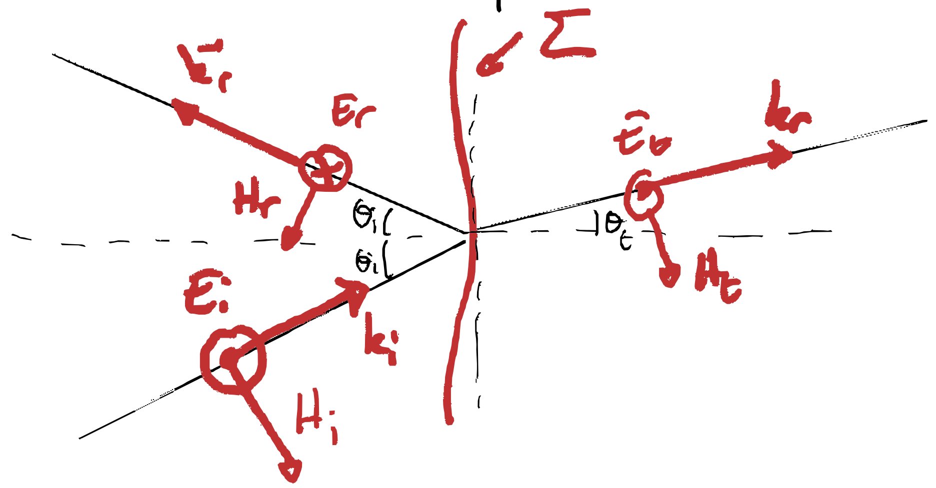
$$1-\frac{n_2^2}{n_1^2} = \left(\frac{n_1^2}{n_2^2}-1\right)\tan^2\theta_i$$

$$\Rightarrow \tan^2\theta_i = \left(\frac{n_2}{n_1}\right)^2$$

$$\Rightarrow \theta_B = \arctan\frac{n_2}{n_1}$$

(Angle de Brewster)
plus de réflexion TM

Polarisation s (transverse électrique)



$$\begin{cases} E_i + E_r = E_t & I \in \Sigma \\ -H_i\cos\theta_i + H_r\cos\theta_i = -H_t\cos\theta_t \end{cases} \Rightarrow \begin{cases} 1 + r_{TE} = t_{TE} \\ k_{iz}(1-r_{TE}) = k_{tz}t_{TE} \end{cases}$$

$$\begin{cases} r_{TE} = \dfrac{k_{iz}-k_{tz}}{k_{iz}+k_{tz}} \\ t_{TE} = \dfrac{2k_{iz}}{k_{iz}+k_{tz}} \end{cases} \Rightarrow \quad r_{TE} = \frac{n_1\cos\theta_i - n_2\cos\theta_t}{n_1\cos\theta_i + n_2\cos\theta_t}$$

$$t_{TE} = \frac{2n_1\cos\theta_i}{n_1\cos\theta_i + n_2\cos\theta_t}$$

II Stabilité cavité laser

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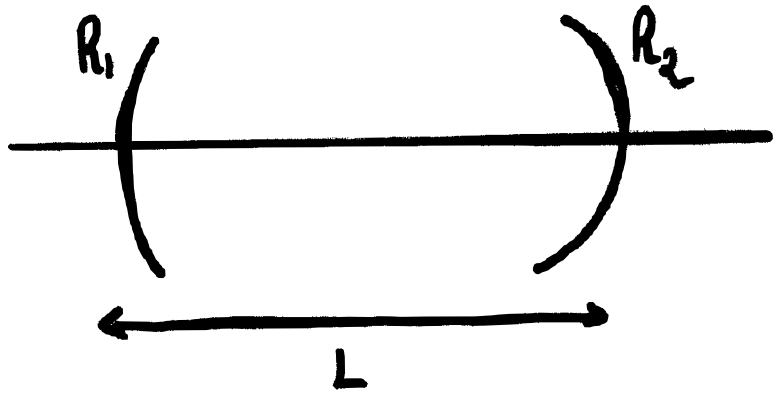
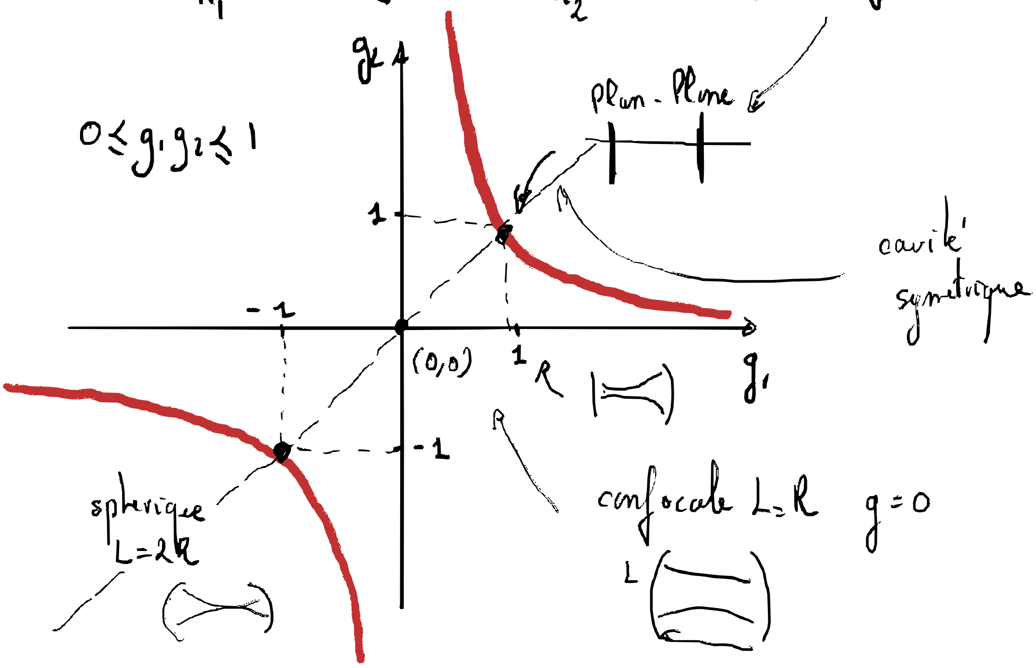


$g_1 = 1 - \frac{L}{R_1}$ et $g_2 = 1 - \frac{L}{R_2}$ si plane $g = 1$



⚠ $0 \leq (1-g)^2 \leq 1 \Rightarrow 0 \leq |1-g| \leq 1$

$$\Rightarrow \begin{cases} 0 \leq (1-g) \leq 1 \\ \text{ou} \\ 0 \leq g-1 \leq 1 \end{cases}$$

III Filtrage externe à la cavité

Principe de Huygens-Fresnel

$a_S(P,t) = \frac{A_0}{SP} \exp\left(j\omega t - jk_0 (SP)\right)$

Principe de Huygens-Fresnel

$$a_S(P,t) = \frac{A_0}{SP}\exp\left(j\omega t - jk_0(SP)\right)$$

$$da_P(M,t) = K\,a_S(P,t)\,\frac{\exp(-jk_0(PM))}{PM}\,\underbrace{t(P)\,d\Sigma}_{d\sigma(P)}$$

$$= K\,\frac{A_0\exp(j\omega t - jk_0(SP))}{SP}\,\frac{\exp(-jk_0(PM))}{PM}\,d\sigma(P)$$

$$da_P(M,t) = \frac{KA_0}{SP}\,\frac{\exp(j\omega t)}{PM}\,\exp\left(-jk_0(SPM)\right)d\sigma(P)$$

à grande distance $SP \simeq SO$ et $PM \simeq OM$ on note $C = \frac{K}{SP\ PM}$ = constante (Diffraction de Fraunhofer)

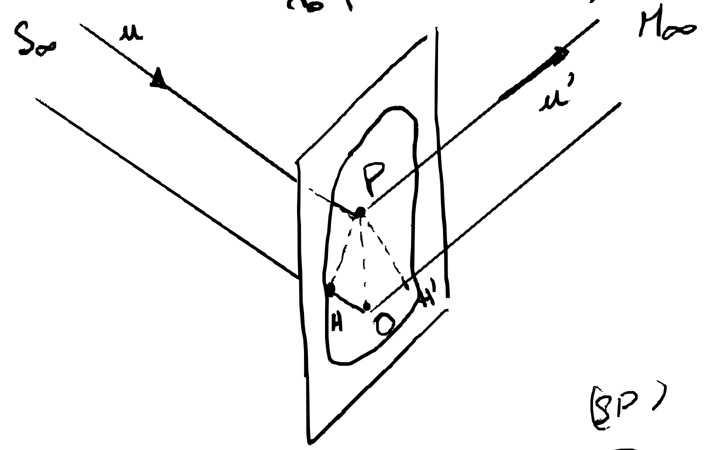
$$da_P(M,t) = C A_0 \exp(j\omega t)\exp\left(-jk_0(SPM)\right)\underbrace{t(P)\,d\Sigma}_{d\sigma(P)}$$

On note $\varphi_A(M) = \underbrace{k_0}_{2\pi/\lambda_0}(SAM)$

$$da_P(M,t) = CA_0\exp(j\omega t)\exp(-j\varphi_P(M))\,t(P)\,d\Sigma$$

$$= CA_0\exp(j\omega t)\exp(-j\varphi_O(M))\exp\Big(-j\underbrace{(\varphi_P(M)-\varphi_O(M))}_{\Delta\varphi_P(M)}\Big)t(P)\,d\Sigma$$

On note $\Delta\varphi_P(M) = \frac{2\pi}{\lambda_0}\left((SPM) - (SOM)\right)$



$$(SPM) - (SOM) = \overbrace{(SP)}^{} + \overbrace{(PM)}^{} - \Big(\overbrace{(SH) + (HO)}^{(SP)} + \overbrace{(OH') + (H'M)}^{(PM)}\Big)$$

$$= -(HO) - (OH') = \vec{u}\cdot\overrightarrow{OP} - \vec{u}'\cdot\overrightarrow{OP}$$

$$= (\vec{u} - \vec{u}')\cdot\overrightarrow{OP}$$

$$= -(HO) - (OH') = \vec{u}.\vec{OP} - \vec{u}'\vec{OP}$$

$$= (\vec{u} - \vec{u}').\vec{OP}$$

On note $a_0(M) = A_0 \exp(-j \varphi_0(M))$

$$da_p(M,t) = C a_0(M) \exp(+j\omega t) \exp\left(j k_0 (\vec{u}' - \vec{u}).\vec{OP}\right) t(P) d\Sigma$$

on note $\vec{u} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$ et $\vec{u}' = \begin{pmatrix} \alpha' \\ \beta' \\ \gamma' \end{pmatrix}$ et $\vec{OP} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$

$$da_p(M,t) = C a_0(M) \exp(j\omega t) \exp\left(j k_0 \left((\alpha'-\alpha)x + (\beta'-\beta)y\right)\right) H(P) d\Sigma$$

$$a(M,t) = \iint_\Sigma da_p(M,t) = C a_0(M) \exp(j\omega t) \iint_\Sigma e^{jk((\alpha'-\alpha)x + (\beta'-\beta)y)} H(x,y) dK$$

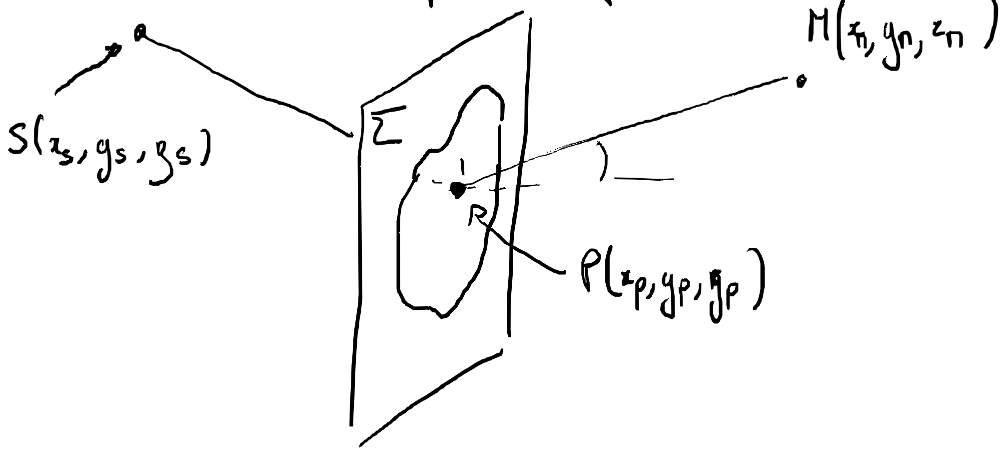
Remarque transformation de Fourier $\mathcal{F}$

$\mathcal{F}[f](k) = \alpha \int dx\, f(x)\, e^{\pm i\gamma kx}$, $\mathcal{F}^{-1}[g](k) = \beta \int dx\, g(x)\, e^{\mp i\gamma kx}$

avec $\gamma = 2\pi\alpha\beta$

$$\text{Or } a(M,t) = C a_0(M) \exp(j\omega t) \iint_\Sigma t(x,y)\, e^{j[(\alpha'-\alpha)x + (\beta'-\beta)y]} dx dy$$

On reconnait une transformée de fourier 2D



$$\alpha' = \frac{x_P - x_S}{\sqrt{(x_P - x_S)^2 + (y_P - y_S)^2 + (z_P - z_S)^2}} \quad \beta' = \frac{y_P - y_S}{\sqrt{(x_P - x_S)^2 + (y_P - y_S)^2 + (z_P - z_S)^2}}$$

$$\alpha' = \frac{x_p - x_s}{\sqrt{(x_p - x_s)^2 + (y_p - y_s)^2 + (z_p - z_s)^2}} \qquad \beta' = \frac{y_p - y_s}{\sqrt{(x_p - x_s)^2 + (y_p - y_s)^2 + (z_p - z_s)^2}}$$

$$\gamma' = \frac{z_p - z_s}{\sqrt{(x_p - x_s)^2 + (y_p - y_s)^2 + (z_p - z_s)^2}}$$

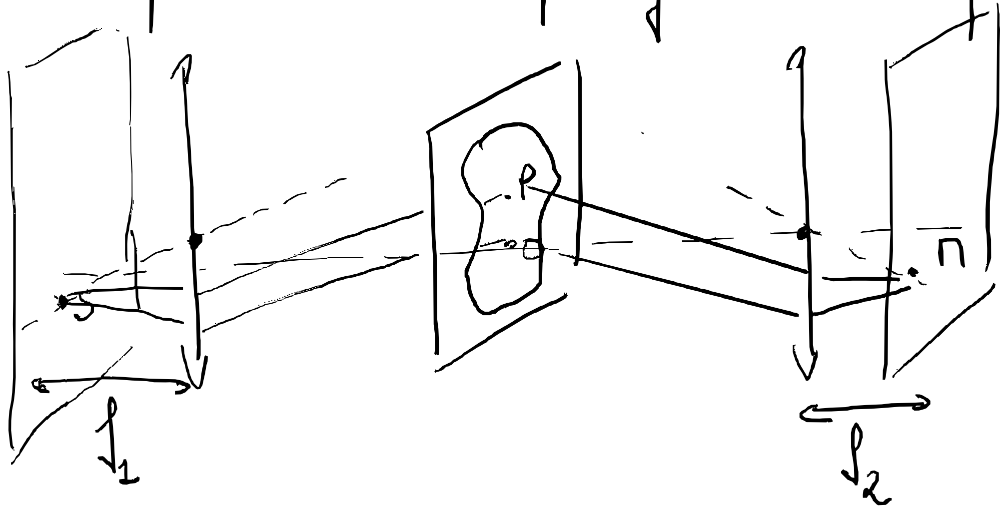
$$\alpha = \frac{x_\Pi - x_p}{\sqrt{(x_\Pi - x_p)^2 + (y_\Pi - y_p)^2 + (z_\Pi - z_p)^2}} \qquad \beta = \frac{y_\Pi - y_p}{\sqrt{(x_\Pi - x_p)^2 + (y_\Pi - y_p)^2 + (z_\Pi - z_p)^2}}$$

$$\gamma = \frac{z_\Pi - z_p}{\sqrt{(x_\Pi - x_p)^2 + (y_\Pi - y_p)^2 + (z_\Pi - z_p)^2}}$$

S et Π proche de l'axe optique et on met $P = (0,0,0)$

$\alpha' \simeq -\frac{x_s}{z_s}$ $\beta' \simeq -\frac{y_s}{z_s}$ $\gamma' \simeq 1$ $\alpha \simeq \frac{x_\Pi}{z_\Pi}$, $\beta \simeq \frac{y_\Pi}{z_\Pi}$; $\gamma \simeq 1$

Si on place S dans le plans focale de 2 lentille



alor $\alpha' = -\frac{x_s}{f_1}$; $\beta' = -\frac{y_s}{f_1}$; $\alpha = \frac{x_\Pi}{f_2}$; $\beta = \frac{y_\Pi}{f_2}$

Quelque exemple de transforme de Fourie

- On definie la distribution de Dirac δ tel que

$$\int \delta(y)\, e^{+i\gamma y x}\, dy = 1$$

Donc que la définition de la transformée de Fourier $\mathcal{F}$

$$\int \delta(y)\, e^{\mp i\gamma y x}\, dy = 1$$

Donc avec la définition de la transformée de fourier $\mathcal{F}$

$$\mathcal{F}[f](k) = \alpha \int f(x) e^{\mp i\gamma k x} dx \;;\; \mathcal{F}^{-1}[f](k) = \beta \int f(x) e^{\pm i\gamma k x} dx$$

avec $\gamma = 2\pi\alpha\beta$

$$\boxed{\begin{array}{ll} \mathcal{F}^{-1}[\delta] : x \mapsto \beta & \text{et } F[x\mapsto\beta] = \delta \\ F[\delta] : x \mapsto \alpha & F^{-1}[x\mapsto\alpha] = \delta \end{array}} \text{ donc } \boxed{\frac{1}{2\pi}\int e^{\mp i y x} dx = \delta(y)}$$

- Convolution et multiplication

$$(f*g)(x) = \int f(x')\, g(x-x')\, dx'$$

$$\mathcal{F}[f*g] : k \mapsto \alpha \int dx \int f(x')\, g(x-x')\, dx'\, e^{\mp i\gamma k x} \underset{\substack{u = x-x' \\ dx = du}}{=} \alpha \int du\, g(u) e^{\mp i\gamma k u} \int f(x') e^{\mp i\gamma k x'} dx'$$

$$= \frac{1}{\alpha} F[f] \cdot F[g]$$

$$\text{Donc } \boxed{\begin{array}{l} F[f*g] = \frac{1}{\alpha} F[f]\cdot F[g] \\ \text{et } F^{-1}[f*g] = \frac{1}{\beta} F^{-1}[f] \cdot F^{-1}[g] \end{array}}$$

$$\mathcal{F}[f] * \mathcal{F}[g] : k \mapsto \int dy\, \alpha \int dx\, f(x)\, e^{\mp i y x}\, \alpha \int dx'\, g(x') e^{\mp i\gamma(k-y)x'}$$

$$= \alpha^2 \int dx\, f(x) \int dx'\, g(x') e^{\mp i\gamma k x'} \underbrace{\int dy\, e^{\mp i\gamma y (x-x')}}_{\frac{2\pi}{\gamma}\delta(x-x')}$$

$$= \alpha \frac{\alpha 2\pi}{\gamma} \int dx\, f(x) g(x) e^{\mp i\gamma k x}$$

$$= \frac{1}{\beta} F[f\cdot g](k)$$

$$\text{Donc } \boxed{\begin{array}{l} F[f\cdot g] = \beta\, F[f] * F[g] \\ F^{-1}[f\cdot g] = \alpha\, F^{-1}[f] * F^{-1}[g] \end{array}}$$

Donc $\boxed{\begin{aligned} F[f\cdot g] &= \beta\, F[f] * F[g] \\ F^{-1}[f\cdot g] &= \alpha\, F^{-1}[f] * F^{-1}[g] \end{aligned}}$

$$\boxed{\begin{aligned} \mathcal{F}[x\mapsto f(x+a)](x) &= \left(x \mapsto e^{\pm i\gamma x a}\,\mathcal{F}[f](x)\right) \\ \mathcal{F}[x\mapsto f(ax)](x) &= \left(x \mapsto \frac{1}{a}\mathcal{F}[f]\left(\frac{x}{a}\right)\right) \end{aligned}}$$

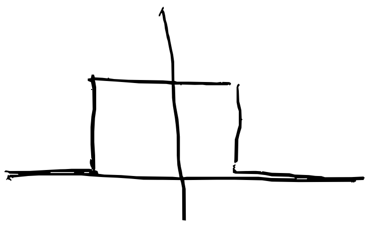
- Peigne de Dirac

$$\text{Ш}_T(x) = \sum_{n\in\mathbb{Z}} \delta(x - Tn) \qquad \mathcal{F}[\text{Ш}_T] = \frac{1}{\beta T}\,\text{Ш}_{1/T}$$

$$\sum_{n\in\mathbb{Z}} e^{i\gamma x n T} = \frac{2\pi}{\gamma T}\sum_{m\in\mathbb{Z}} \delta\left(k - \frac{2\pi m}{\gamma T}\right) \leftarrow \text{Poisson}$$

- Porte

$$\Pi(x) = \begin{cases} 0 & \text{si } x^2 > 1 \\ \frac{1}{2} & \text{si } x^2 = 1 \\ 1 & \text{si } x^2 < 1 \end{cases}$$



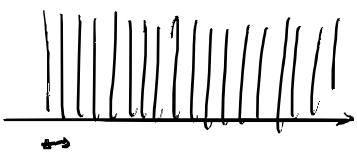
$$\mathcal{F}[\Pi](k) = \alpha\int_{-1}^{1} e^{\mp i\gamma k x}\,dx = \frac{\alpha}{\mp i\gamma k}\left(e^{\mp i\gamma k} - e^{\pm i\gamma k}\right)$$

$$= \alpha 2 \operatorname{sinc}(\gamma k)$$

$\mathcal{F}[\Pi](x) = 2\alpha \operatorname{sinc}(\gamma x)$ avec $\operatorname{sinc}(x) = \frac{\sin(x)}{x}$

$$\mathcal{F}\left[x\mapsto \Pi\left(\frac{x}{l}\right)\right](x) = l\, 2\alpha \sin(\gamma l x)$$

Diaphragme composé d'une infinité de fente infiniment fines



$$f_1(x) = \text{Ш}_a(x) = \frac{1}{a}\text{Ш}\left(\frac{x}{a}\right)$$

$\downarrow TF \qquad \downarrow TF$

$$\frac{1}{a\beta}\,\Pi_{1/a\beta}(x) = \alpha\,\underbrace{\text{Ш}(a\alpha\beta x)}_{\tilde{f}_1(x)}$$

$$\delta\left(\frac{x}{a} - n\right) = a\,\delta(x - an)$$

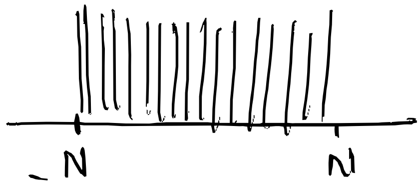
$$\frac{1}{\alpha\beta}\,\text{Ш}_{1/\alpha\beta}(x) = \underbrace{\alpha\,\text{Ш}(\alpha\beta x)}_{\tilde{f}_1(x)}$$

$$\delta\left(\frac{x}{a}-n\right) = a\,\delta(x-an)$$

$$\mathcal{F}[\text{Ш}](x) = \frac{1}{\beta}\sum_m \delta\left(x-\frac{m}{\alpha\beta}\right) \qquad \mathcal{F}\left[x\mapsto \frac{1}{a}\underbrace{\text{Ш}\left(\frac{x}{a}\right)}_{\text{Ш}_a(x)}\right] = \mathcal{F}[\text{Ш}](ax) = \alpha\,\text{Ш}(a\alpha\beta x)$$

$$= \frac{1}{\beta}\sum \alpha\beta\,\delta(\alpha\beta x - m) = \alpha\,\text{Ш}(\alpha\beta x) \qquad = \frac{\alpha}{a\alpha\beta}\,\text{Ш}_{1/a\alpha\beta}(x)$$

Nombre fini de fente infiniment fine



$$f_2(x) = \frac{1}{a}\,\text{Ш}\left(\frac{x}{a}\right)\cdot\Pi\left(\frac{x}{(N+\frac{1}{2})a}\right)$$

$$\mathcal{F}\left[x\mapsto \Pi\left(\frac{x}{(N+\frac{1}{2})a}\right)\right](x) = (2N+1)a\alpha\,\operatorname{sinc}\left(\gamma\left(N+\frac{1}{2}\right)ax\right)$$

$$f_2(x) \xrightarrow{TF} \beta\,\tilde{f}_1 * \left(x\mapsto (2N+1)a\alpha\,\operatorname{sinc}\left(\gamma\left(N+\frac{1}{2}\right)ax\right)\right)$$

$$\alpha\,\text{Ш}(a\alpha\beta x) = \alpha\sum_{n\in\mathbb{Z}}\delta(a\alpha\beta x - n) = \frac{1}{a\beta}\sum_{n\in\mathbb{Z}}\delta\left(x-\frac{n}{a\alpha\beta}\right)$$

$$= (2N+1)\alpha\sum_{n\in\mathbb{Z}}\operatorname{sinc}\left(\gamma\left(N+\frac{1}{2}\right)a\left(x-\frac{n}{a\alpha\beta}\right)\right)$$

$$\tilde{f}_2(x) = (2N+1)\alpha\sum_{n\in\mathbb{Z}}\operatorname{sinc}\left(\gamma\left(N-\frac{1}{2}\right)\left(ax-\frac{n}{\alpha\beta}\right)\right)$$

or aussi $f_2(x) = \sum_{n=-N}^{n=N}\delta(x-an)$

donc $\tilde{f}_2(x) = \alpha\sum_{n=-N}^{N} e^{\pm i\gamma n x a}$

(somme géométrique Cas $q\neq 1$: $\sum^{a} q^n = q^b\,\frac{1-q^{a-b+1}}{\cdots}$

donc $\tilde{f}(x) = \alpha \sum_{n=-N} e^{\pm i \gamma n x a}$

somme geometrique Cas $q \neq 1$: $\sum_{n=b}^{a} q^n = q^b \frac{1-q^{a-b+1}}{1-q}$

Cas $q = 1$ $\sum_{n=b}^{a} 1 = a-b+1$ $= \frac{q^{a+1} - q^b}{q-1}$

$$\tilde{f}(x) = \alpha \sum_{n=-N}^{N} \left(e^{\pm i \gamma x a}\right)^n = \alpha \frac{e^{\mp i \gamma x a N} - e^{\pm i \gamma x a (N+1)}}{1 - e^{\pm i \gamma x a}}$$

$$= \alpha e^{\mp i \gamma x N} \frac{1 - e^{\pm i \gamma x a (2N+1)}}{1 - e^{\pm i \gamma x a}}$$

$$= \alpha e^{\mp i \gamma x N} \underbrace{\frac{e^{\pm i \gamma x a (2N+1)/2}}{e^{\pm i \gamma x a/2}}}_{1} \frac{e^{\mp i \frac{\gamma}{2} x a (2N+1)} - e^{\pm i \frac{\gamma}{2} x a (2N+1)}}{e^{\mp i \frac{\gamma}{2} x a} - e^{\pm i \frac{\gamma}{2} x a}}$$

$$= \alpha \frac{\sin\left((2N+1)\frac{\gamma}{2} a x\right)}{\sin\left(\frac{\gamma}{2} a x\right)}$$

Donc $\tilde{f}(x) = \alpha (2N+1) \sum_{n \in \mathbb{Z}} \mathrm{sinc}\left(\frac{\gamma}{2}(2N+1)\left(ax - \frac{n}{\alpha\beta}\right)\right)$

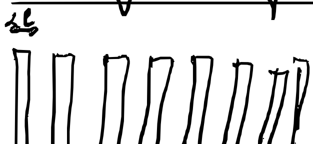
$$= \alpha \frac{\sin\left(\frac{\gamma}{2}(2N+1) a x\right)}{\sin\left(\frac{\gamma}{2} a x\right)}$$

Demontrons pas la meme occasion

$$\sum_{n \in \mathbb{Z}} \mathrm{sinc}\left(\pi (2N+1)(a\alpha\beta - n)\right) = \frac{1}{(2N+1)} \frac{\sin\left(\frac{\gamma}{2}(2N+1) a x\right)}{\sin\left(\frac{\gamma}{2} a x\right)}$$

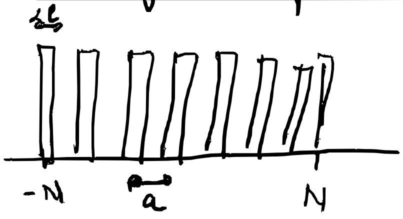
avec $N \in \mathbb{Z}$, $(\gamma = 2\pi\alpha\beta, a) \in \mathbb{R}^2$

Nombre fini de fentes de dimension finie



$f(x) = \sum_{n=-N}^{N} \Pi\left(\frac{x-na}{\ell}\right)$

Nombre fini de fentes de dimension finie



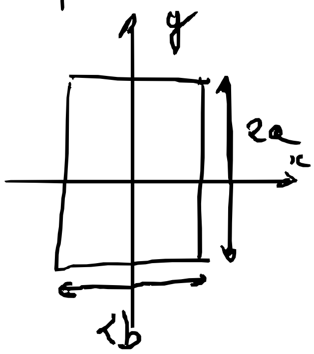
$$f_3(x) = \sum_{n=-N}^{N} \Pi\left(\frac{x-na}{\ell}\right)$$

$$= \left(f_2 * \left(x \mapsto \Pi\left(\frac{x}{\ell}\right)\right)\right)(x)$$

$$\tilde{f}_3(x) = \frac{1}{\alpha}\, \tilde{f}_2(x) \cdot \ell 2\alpha \,\mathrm{sinc}(\gamma \ell x) = 2\ell\, \tilde{f}_2(x) \cdot \mathrm{sinc}(\gamma \ell x)$$

$$= 2\ell\alpha \frac{\sin\left(\frac{\gamma}{2}(2N+1)ax\right)}{\sin\left(\frac{\gamma}{2}ax\right)}$$

Diffraction d'un rectangle



$$f(x,y) = \Pi\left(\frac{x}{a}\right)\Pi\left(\frac{y}{b}\right)$$

$$\tilde{f}(x,y) = \alpha^2 4ab \,\mathrm{sinc}(\gamma a x)\,\mathrm{sinc}(\gamma b y)$$

Diffraction d'une Pupille circulaire

$$\vec{x} = (x,y) \qquad f(\vec{x}) = \Pi\left(\frac{\|\vec{x}\|_2}{D/2}\right)$$

$$\tilde{f}(\theta,\psi) = \alpha^2 \iint f(x,y)\, e^{\mp i\gamma(\theta x + \psi y)}\, dx\, dy$$

$r = \|x\|_2 = \sqrt{x^2+y^2}$

$$= \alpha^2 \int_0^{\infty} r\, \Pi\left(\frac{r}{D/2}\right) \int_{-\pi}^{\pi} e^{\mp i\gamma \rho r \cos\varphi}\, d\varphi\, dr$$

$(x,y) \mapsto (r,\varphi)$

avec $\rho = \sqrt{\theta^2+\psi^2}$

$$J_0(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-iz\cos\varphi}\, d\varphi$$

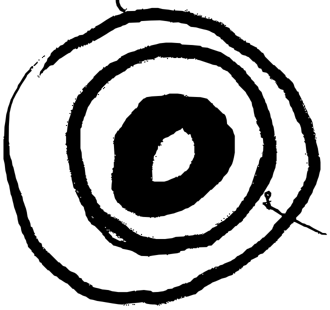
$$J_1(z) = \frac{1}{i\pi}\int_{-\pi}^{\pi} e^{\mp iz\cos\varphi}\cos\varphi\, d\varphi \quad \text{et} \quad \int_0^{x} y\, J_0(y)\, dy = x J_1(x)$$

avec $\rho = \sqrt{\theta^2+\psi^2}$ $\quad 2\pi \int_{\pi}$

$$J_1(z) = \frac{1}{i\pi}\int_{-\pi}^{\pi} e^{\mp i z \cos\varphi} \cos\varphi \, d\varphi \quad \text{et} \quad \int_0^{x} y\, J_0(y)\, dy = x J_1(x)$$

$$f(\theta,\psi) = \frac{2\pi a^2}{\rho^2 \gamma^2} \int^{\gamma \frac{D}{2}\rho} y\, J_0(y)\, dy$$

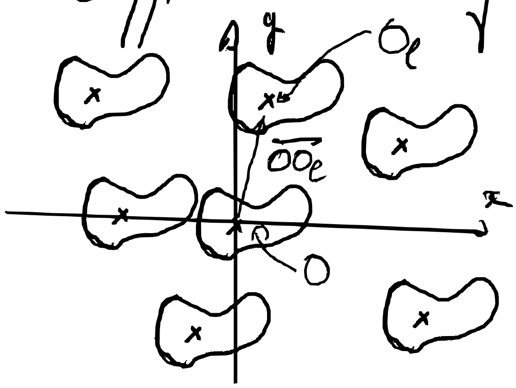
$$= \frac{2\pi a^2}{\rho^2\gamma^2}\, \gamma\frac{D}{2}\rho\, J_1\left(\gamma\frac{D}{2}\rho\right) = \underbrace{\pi a^2 \frac{D^2}{4}}_{S} \left[\frac{2 J_1\left(\gamma \frac{D}{2}\rho\right)}{\gamma\frac{D}{2}\rho}\right]$$



Bessel

tache d'Airy

Diffraction de motifs identique



$$f_{tot}(\overrightarrow{OP}) = \sum_{1\le \ell\le N} f(\overrightarrow{OP} - \overrightarrow{OO_\ell})$$

$$\tilde{t}_{top}(\vec{k}) = a^2 \iint f_{tot}(\overrightarrow{OP})\, e^{\mp i\gamma \vec{k}\cdot\overrightarrow{OP}}\, dS_P$$

$$= a^2 \sum_{1\le \ell\le N} \iint f(\overrightarrow{OP} - \overrightarrow{OO_\ell})\, e^{\mp i\gamma \vec{k}\overrightarrow{OP}}\, dS_P$$

$\overrightarrow{OP'} = \overrightarrow{OP} - \overrightarrow{OO_\ell}$

$$= \sum_{1\le \ell\le N} e^{\mp i\gamma\vec{k}\cdot\overrightarrow{OO_\ell}}\, \underbrace{a^2 \iint f(\overrightarrow{OP'})\, e^{\mp i\gamma \vec{k}\overrightarrow{OP'}}\, dS_{P'}}_{\tilde{f}(k)}$$

Champ de dirac $\star$ porte forme convolute

$$f_{tot}(\overrightarrow{OP}) = g(\overrightarrow{OP}) \star f(\overrightarrow{OP})$$

$$f_{tot}(\vec{OP}) = g(\vec{OP}) * f(\vec{OP})$$

$$\left(\vec{OP} \mapsto \sum_{1\leq l\leq N} \delta(\vec{OP} - \vec{OO_l})\right) * f$$

or $F\left[\vec{x} \mapsto (f*g)(\vec{x})\right] = \frac{1}{\alpha^2} F[f] \cdot F[g]$

dim = 2

$$F\left[\vec{x} \mapsto \delta(\vec{x})\right] = \alpha^2$$

$$F\left[\vec{x} \mapsto f(\vec{x}+\vec{a})\right] = \left(\vec{x} \mapsto e^{\pm i \gamma \vec{x}\vec{a}} F[f](\vec{x})\right)$$

donc $F\left[\vec{x} \mapsto \delta(\vec{x} - \vec{a})\right] = \left(x \mapsto \alpha^2 e^{\mp i \gamma \vec{x}\vec{a}}\right)$

donc $F\left[\vec{x} \mapsto \sum_{1\leq l\leq N} \delta(\vec{x} - \vec{OO_l})\right](\vec{x}) = \alpha^2 \sum_{1\leq l\leq N} e^{\mp i \gamma \vec{x}\vec{OO_l}}$

donc $\tilde{f}_{tot} = \left(\vec{x} \mapsto \sum_{1\leq l\leq N} e^{\mp i \gamma \vec{x}\vec{OO_l}}\right) \cdot \tilde{f}$

<u>Pour des motif aléatoire</u>

Ce que l'on mesure c'est $|\tilde{f}_{tot}|^2$

et $\left\| \sum_{1\leq l\leq N} e^{\mp i \gamma \vec{x}\vec{OO_l}} \right\|^2 = \sum_{1\leq l,m\leq N} e^{\mp i \gamma \vec{x} \vec{O_m O_l}}$

Pour $m = l$ $e^{\mp i \gamma \vec{x} \vec{O_m O_l}} = 1$; N terme

et les reste est accompagné de son conjugué

$$= N + 2 \underbrace{\sum_{1\leq l<m\leq} \cos(\gamma \vec{x} \vec{O_m O_l})}_{N^2 - N \text{ terme de moyenne nulle}}$$

$$= N + 2 \underbrace{\sum_{1 \leq l < m \leq} \cos(\gamma x \, \theta_m \theta_l)}_{N^2 - N \text{ terme de moyenne nulle}}$$

$$\mathcal{E}_N(M) = N \, \mathcal{E}_{motif}(N)$$

IV speckles

Coherence du laser.

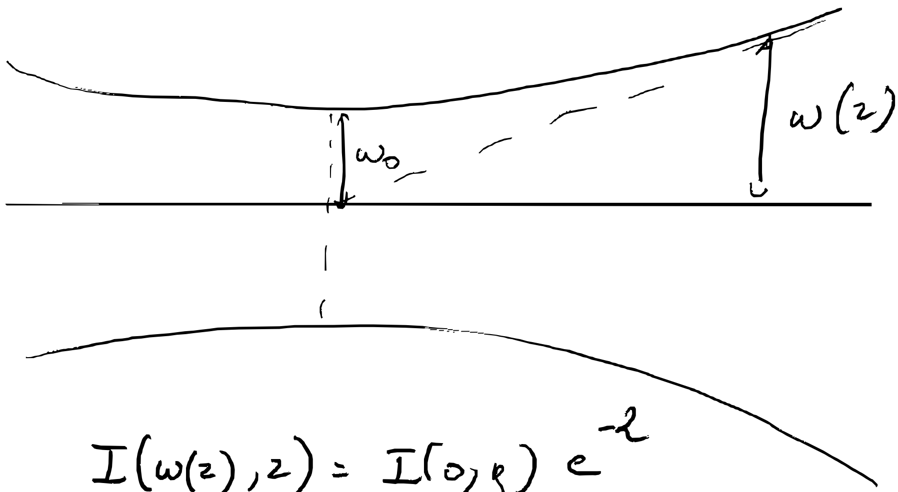
- Chaque petit point de la surface diffuse une onde
- les chemins optiques sont tous légèrement différents
- les phases deviennent aléatoires
- toutes ces ondes interfèrent

V waist

$\omega(z)$ rayon du faisceau

$$I(r,z) = I_0 \left(\frac{w_0}{w(z)}\right)^2 e^{-2r^2/w(z)^2}$$

distance à l'axe



$$I(w(z),z) = I(0,z) \, e^{-2}$$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} \qquad z_R = \frac{\pi w_0^2}{\lambda}$$

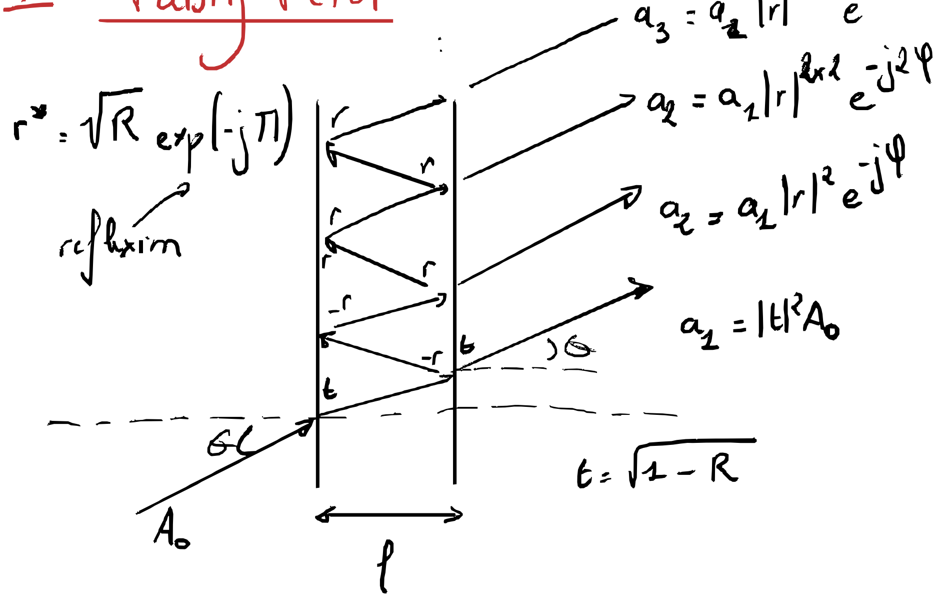
$$w^2(z) = w_0^2 + w_0^2 \, z^2$$

$$w(z) = w_0 \sqrt{1+\left(\frac{z}{z_R}\right)^2} \qquad z_R = \frac{\quad}{\lambda}$$

$$w^2(z) = w_0^2 + \frac{w_0^2}{z_R^2} z^2$$

- affine en z^2

VI Fabry Perot



La difference de marche entre a_{n+1} et a_n est

$$\delta = 2\ell \cos(\theta)$$

donc la difference de phase est

$$\varphi = \frac{2\pi}{\lambda_0} 2\ell\cos\theta = \frac{4\pi \ell \cos\theta}{\lambda_0}$$

$$a_{n+1} = a_n R e^{-j\varphi} = (Re^{-j\varphi})^n a_1 = (1-R)A_0 \left(R\exp(-j\varphi)\right)^n$$

$$a = \sum_{k=1}^{+\infty} a_k = (1-R)A_0 \sum_{k=1}^{+\infty} (Re^{-j\varphi})^{k-1} = \frac{(1-R)A_0}{1-R\exp(-j\varphi)}$$

$$|q|<1 \rightarrow S_n = \sum_{0\leq k\leq n} q^k = \frac{1-q^{n-1}}{1-q}$$

$$a = \frac{(1-R)A_0}{1-R\cos\varphi + jR\sin\varphi} \rightsquigarrow |a|^2 = \frac{(1-R)^2 A_0^2}{(1-R\cos\varphi)^2 + R^2\sin^2\varphi}$$

E_0^2

$$T = \frac{E}{\quad} = \frac{(1-R)^2}{\quad} = \frac{(1-R)^2}{(1-R)^2 \ldots}$$

$$\frac{\quad}{1 - R\cos\varphi + jR\sin\varphi} \qquad \text{par} \qquad \frac{\quad}{(1-R\cos\varphi)^2 + R^2\sin^2\varphi}$$

$$\Upsilon = \frac{\mathcal{E}}{\mathcal{E}_0} = \frac{(1-R)^2}{1+R^2-2R\cos\varphi} = \frac{(1-R)^2}{(1-R)^2 - 4R\sin^2\frac{\varphi}{2}}$$

$$\cos^2\varphi + \sin^2\varphi = 1$$
$$\cos 2\varphi = \cos^2\varphi - \sin^2\varphi = 1 - 2\sin^2\varphi$$

$$= \frac{1}{1 + \frac{4R}{(1-R)^2}\sin^2\frac{\varphi}{2}}$$

Donc $\Upsilon = 1 \Leftrightarrow \sin^2\varphi/2 = 0$

$$\Leftrightarrow \frac{\varphi}{2} = q\pi$$

$$\Leftrightarrow \frac{2n l\cos\theta}{d_0} = q\pi \quad q \in \mathbb{Z}$$

$$\Leftrightarrow 2l\cos\theta = q d_0 \quad q \in \mathbb{Z}$$

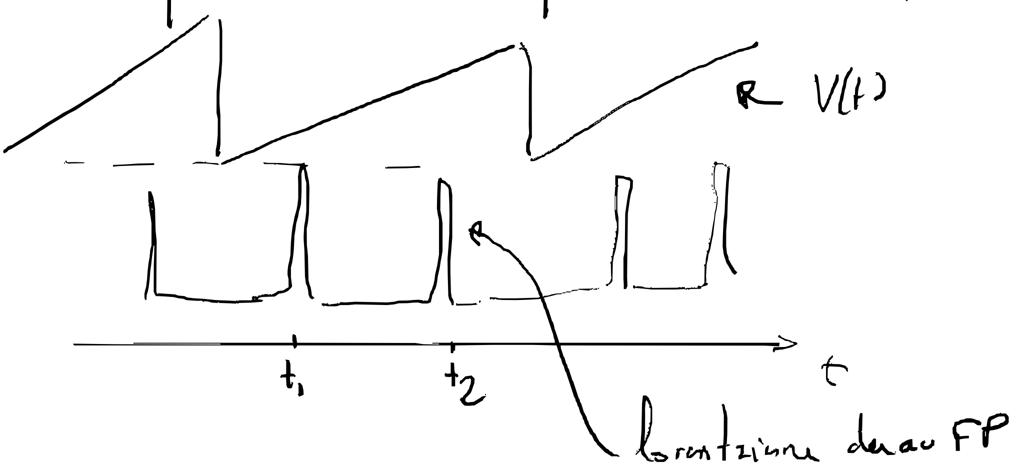
ici $l(t) = b + at$ avec une fréquence $\frac{1}{T}$

Si le laser est monochromatique

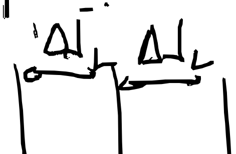
à t_1 : $2l(t_1)\cos\theta = q_1 d_0$

à t_2 : $2l(t_2)\cos\theta = q_2 d_0$

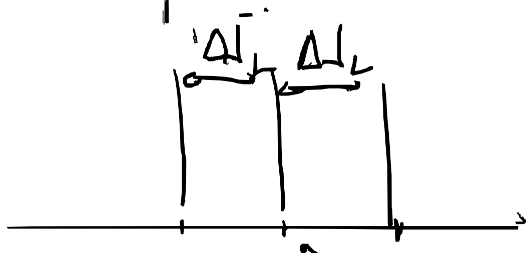
ce que l'on voit théoriquement à l'oscilloscope



Mais le spectre du laser a une structure



Mais le spectre du laser a une structure



$d_{p_1}' = \frac{2L}{p_1}$ $\quad$ $d_{p_2} = \frac{2L}{p_2}$

$p_2 = p_1 + 1$

Je note $d_0 = \frac{(d_{p_1} + d_{p_2})}{2}$ et $\Delta d_L = d_{p_1} - d_{p_2}$

$d_{p_1} = d_0 + \frac{\Delta d_L}{2}$ et $d_{p_2} = d_0 - \frac{\Delta d_L}{2}$

$\nu_p = p\frac{c}{2L}$ $\quad$ $\Delta \nu_L = \frac{c}{2L}$ et $\nu_0 = \frac{c}{d_0}$

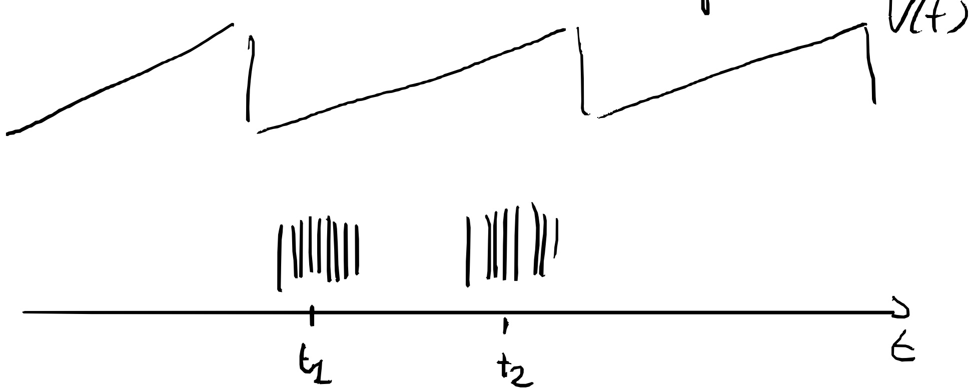
$d_0 \pm \frac{\Delta d_L}{2} = \frac{c}{\nu_0 \mp \frac{\Delta \nu_L}{2}}$ donc

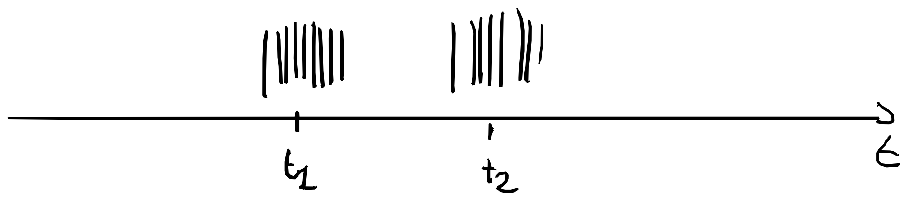
$$\Delta d_L = \mp 2 d_0 \pm \frac{2c}{\nu_0 \mp \frac{\Delta \nu_L}{2}} = \mp 2c\left(\frac{1}{\nu_0} - \frac{1}{\nu_0 \mp \frac{\Delta \nu_L}{2}}\right)$$

$$= \mp \frac{2c}{\nu_0}\left(1 - \frac{1}{1 \mp \frac{\Delta \nu_L}{\nu_0 2}}\right) \simeq \mp \frac{2c}{\nu_0}\left(1 - 1 \mp \frac{\Delta \nu}{\nu_0 2}\right)$$

$$\boxed{\Delta d_L \simeq \frac{\cancel{2}c}{\nu_0^2}\Delta \nu_L = \frac{d_0}{\nu_0}\Delta \nu_L = \frac{d_0}{\nu_0}\frac{c}{2L}}$$

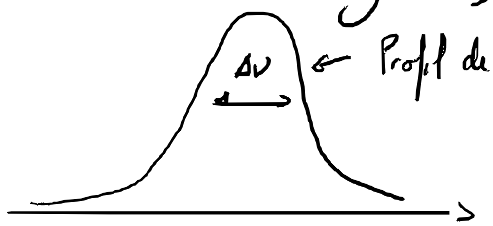
Donc si on voit sur l'oscilloscope





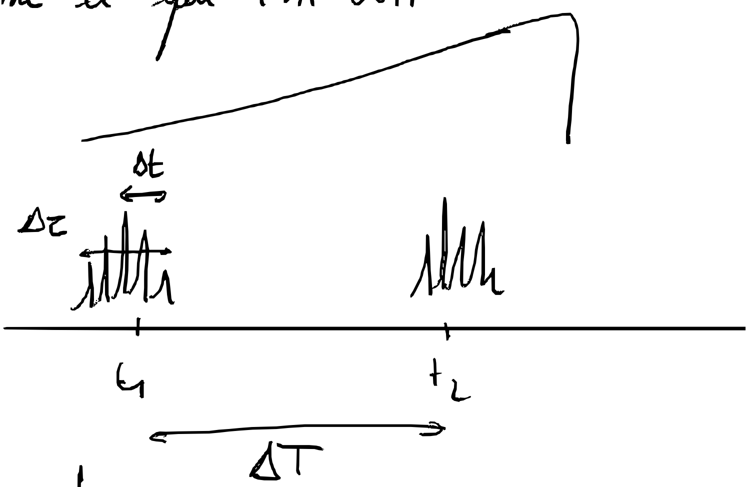
Mais Pas touts les λ sont amplifier pareil

Dans la courbe de gain ;



← Profil de raies d'émission

Donc ce que l'on voit



Correspondre

ΔT → frequence du fabripero 10 GHz

Δt → $\Delta\nu_L$ à trouver $\Delta\lambda_L = \frac{\lambda_0}{\nu_0} \Delta\nu_L$

$\Delta\tau$ → $\Delta\nu_L$

Il faut pas faire un produit en croix